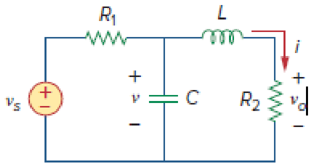
Chapter 16, Problem 10PP

Problem

Obtain the state variable model for the circuit shown in Fig. 16.23. Let *R*1 = 1, *R*2 = 2, *C* = 0.5, and *L* = 0.2 and obtain the transfer function.

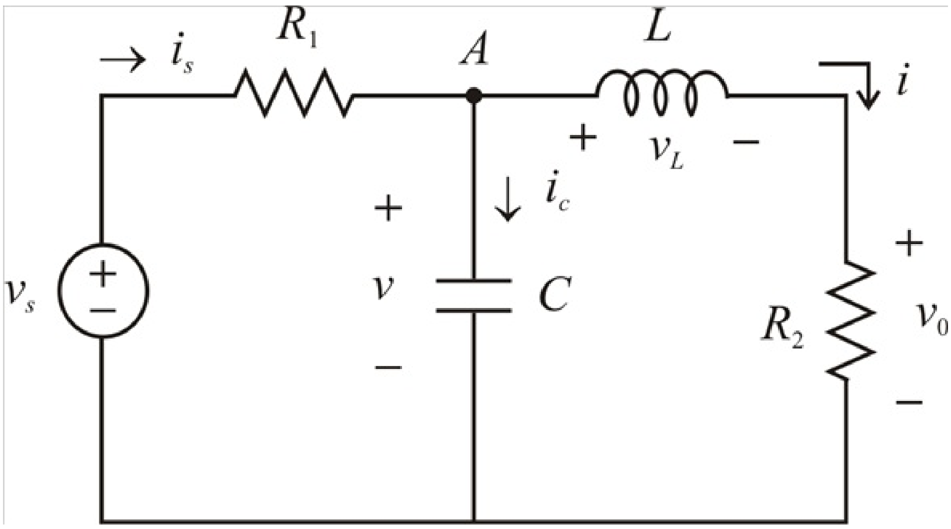
Figure 16.23:



Step-by-step solution

Step 1 of 5

Given circuit



Step 2 of 5

Consider the inductor current *i* and capacitor voltage *v* as the state variables.

$$v_L = L\frac{di}{dt}$$

$$i_c = C\frac{dv}{dt}$$

Applying KCL at node *A* gives

$$i_s = i_c + i$$

$$\frac{v_s - v}{R_1} = C\frac{dv}{dt} + i$$

$$C\frac{dv}{dt} = \frac{v_L}{R_1} - \frac{v}{R_1} - i$$

$$\dot{v} = \frac{v_s}{R_1 C} - \frac{v}{R_1 C} - \frac{i}{C} \qquad ---(1)$$

Step 3 of 5

Applying KVL around the second inner loop, we get

$$-v+v_L+v_0=0$$

$$v-v_0=v_L$$

$$v-v_0=v_L$$

$$L\frac{di}{dt}=v-iR_2$$

$$\dot{i}=\frac{v}{L}-\frac{iR_2}{L} \qquad ---(2)$$

And

$$v_0=iR_2$$

From equations (1) and (2) we can constitute the state equations. the output voltage

$$v_0=iR_2 \qquad ---(3)$$

$$\boxed{\begin{aligned}\begin{bmatrix}\dot{v}\\ \dot{i}\end{bmatrix}&=\begin{bmatrix}\frac{-1}{R_1C} & \frac{-1}{C}\\ \frac{1}{L} & \frac{-R_2}{L}\end{bmatrix}\begin{bmatrix}v\\ i\end{bmatrix}+\begin{bmatrix}\frac{1}{R_1C}\\ 0\end{bmatrix}v_s \qquad ---(4)\\ v_0&=\begin{bmatrix}0 & R_2\end{bmatrix}\begin{bmatrix}v\\ i\end{bmatrix} \qquad ---(5)\end{aligned}}$$

Step 4 of 5

Given $R_1=1, R_2=2, C=0.5$ and $L=0.2$ Substituting in equation (4) and (5), where the matrices are

$$A=\begin{bmatrix}\frac{-1}{R_1C} & \frac{-1}{C}\\ \frac{1}{L} & \frac{-R_2}{L}\end{bmatrix}$$

$$A=\begin{bmatrix}\frac{-1}{(1)(0.5)} & \frac{-1}{0.5}\\ \frac{1}{0.2} & \frac{-2}{0.2}\end{bmatrix}$$

$$A=\begin{bmatrix}-2 & -2\\ 5 & -10\end{bmatrix}$$

$$B=\begin{bmatrix}\frac{1}{R_1C}\\ 0\end{bmatrix}$$

$$B=\begin{bmatrix}\frac{1}{(1)(0.5)}\\ 0\end{bmatrix}$$

$$B=\begin{bmatrix}2\\ 0\end{bmatrix}$$

Step 5 of 5

$$C=\begin{bmatrix}0 & R_2\end{bmatrix}$$

$$C=\begin{bmatrix}0 & 2\end{bmatrix}$$

$$sI-A=\begin{bmatrix}s & 0\\ 0 & s\end{bmatrix}-\begin{bmatrix}-2 & -2\\ 5 & -10\end{bmatrix}$$

$$sI-A=\begin{bmatrix}s+2 & 2\\ -5 & s+10\end{bmatrix}$$

The inverse of the above matrices is

$$(sI-A)^{-1}=\frac{\text{adjoint of } A}{\text{determinant of } A}$$

$$(sI-A)^{-1}=\frac{\begin{bmatrix}s+10 & -2\\ +5 & s+2\end{bmatrix}}{s^2+12s+20+10}$$

$$(sI-A)^{-1}=\frac{\begin{bmatrix}s+10 & -2\\ +5 & s+2\end{bmatrix}}{s^2+12s+30}$$

Thus, the transfer function is given by

$$H(s)=C(sI-A)^{-1}B$$

$$H(s)=\frac{\begin{bmatrix}0 & 2\end{bmatrix}\begin{bmatrix}s+10 & -2\\ +5 & s+2\end{bmatrix}\begin{bmatrix}2\\ 0\end{bmatrix}}{s^2+12s+30}$$

$$H(s)=\frac{\begin{bmatrix}0 & 2\end{bmatrix}\begin{bmatrix}2s+20\\ +10\end{bmatrix}}{s^2+12s+30}$$

$$\boxed{\therefore H(s)=\frac{20}{s^2+12s+30}}$$